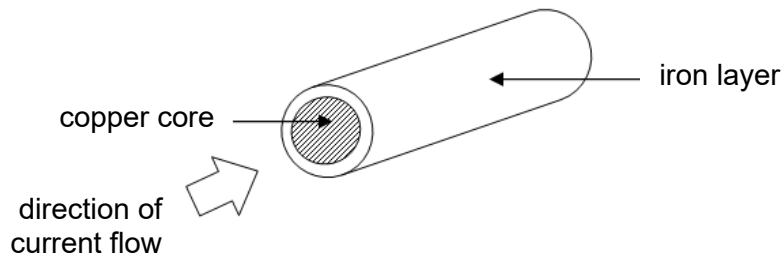


- 22** A resistor is made of a copper cylindrical core of cross-sectional area $2.0 \times 10^{-3} \text{ m}^2$ which is surrounded by iron of cross-sectional area $1.0 \times 10^{-3} \text{ m}^2$.



The resistivities of copper and iron are $1.7 \times 10^{-8} \Omega \text{ m}$ and $1.0 \times 10^{-7} \Omega \text{ m}$ respectively.

What is the resistance per unit length of such a resistor?

- A** $7.8 \times 10^{-6} \Omega \text{ m}^{-1}$
- B** $8.5 \times 10^{-6} \Omega \text{ m}^{-1}$
- C** $8.3 \times 10^{-5} \Omega \text{ m}^{-1}$
- D** $9.2 \times 10^{-5} \Omega \text{ m}^{-1}$

- 23** In each arrangement of resistors, the ammeter has a resistance of 2Ω .